ODD+D Appendix for Comparing Legislative Mechanism Bundles

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This appendix documents the congressional institutional simulator using the ODD and ODD+D model-description conventions [Grimm et al. 2010; Müller et al. 2013]. It describes entities, state variables, scheduling, adaptation, input data, submodels, outputs, empirical boundary checks, assumptions, and reproducibility commands for the simulator used in *A Reproducible Simulation Framework for Comparing Legislative Mechanism Bundles*.

CCS Concepts: • **Computing methodologies** → **Modeling and simulation**; • **Human-centered computing** → *Collaborative and social computing theory, concepts and paradigms.*

Additional Key Words and Phrases: ODD+D, legislative simulation, institutional design, model documentation

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1 Appendix Roadmap

Table 1. Where to look for common reviewer questions.

Reviewer concern	Where to look
What assumptions generate the synthetic worlds?	Table 2 and Section 6
What is empirically checked?	Table 3 and Section 7
What weights enter the vote kernel?	Table 4
What are the selected main-campaign results?	Table 5; full CSV in `reports/simulation-campaign-v21-paper.csv`
What robustness probes exist?	Tables 6, 7, 8, and 9
What are the main limitations?	Section 11
How is the artifact reproduced?	Section 12

2 Purpose

The model compares legislative institutional designs under shared generated political worlds. Its purpose is to stress-test mechanisms that alter productivity, revision moderation, representative responsiveness, public-benefit alignment, minority protection, and resistance to capture. It is not a forecast of a specific Congress. It is a mechanism-comparison simulator: the same generated legislators and bills are routed through many institutional processes so differences can be attributed to procedural rules and incentive structures.

3 Entities, State Variables, and Scales

3.1 Legislators

Legislators have an identifier, party label, ideology, revision-moderation preference, party loyalty, constituency sensitivity, lobbying susceptibility, reputation sensitivity, district preference, district intensity, affected-group sensitivity, and a representation profile. The representation profile records chamber eligibility, represented population share, district

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magnitude, region label, elected/appointed mode, lower- and upper-house term lengths, renewal cohort, renewal cycle, next renewal round, and selector body. These values determine how the legislator weighs ideology, party pressure, constituent/public support, lobbying pressure, revision-moderation incentives, reputation, and concentrated harm before casting a final yes-or-no vote.

3.2 Bills

Bills have an identifier, title, proposer, proposer ideology, original and current policy position, public support, generated public benefit, public-benefit uncertainty, lobby pressure, private gain, salience, issue domain, anti-lobbying-reform flag, affected group, affected-group support, concentrated harm, compensation cost, cosponsor counts, amendment movement, lobbying channel spend, challenge status, citizen-review signals, active-law signals, and alternative-selection signals.

3.3 Lobby Groups

Lobby groups are explicit collective actors. Each group has an identifier, primary issue domain, issue-preference map, preferred policy position, budget, influence intensity, defensive multiplier, information bias, public-campaign skill, capture strategy, and public-support mismatch tolerance. Groups can spend through direct pressure, agenda-access pressure, information distortion, public campaigns, litigation or delay threats, and defensive anti-reform pressure.

3.4 Institutional Processes

An institutional process is a composable module. Processes can screen proposal access, apply leadership agenda control, modify public signals, spend lobbying or attention resources, route bills to procedural tracks, mediate amendments, evaluate affected-group harm, select among alternatives, conduct chamber votes, revise through conference committees, apply vetoes, apply judicial review, route through citizen initiatives or citizen agenda petitions, register enacted laws, or review laws later. The chamber-structure supplement adds chamber-specification, chamber-origin routing, second-chamber amendment and veto powers, last-offer bargaining, lower-house override, committee-assignment configuration, committee-role power, eligibility and retention filters, and ex ante review through insulated fiscal, electoral, audit, ethics, or legal-review bodies. The simulator also includes bounded linked modules for open floor calendars, discharge-petition committee bypass, district-population public signals, pairwise amendment tournaments, long-horizon proposer strategy, omnibus bargaining, campaign-finance influence systems, and constitutional-court architecture. The final decision still resolves to a yes-or-no outcome, enactment or rejection, and a post-bill status quo. Portfolio variants combine low-risk fast passage, middle-risk pairwise alternatives and mediation, high-risk citizen and harm review, proposal bonds, anti-capture safeguards, law-registry correction, and, in the expanded variant, public-opinion, influence-system, omnibus, and court-review proxies.

4 Process Overview and Scheduling

Each run follows a fixed schedule:

(1) Generate a world containing legislators, party positions, lobby groups, a status quo, and a bill stream.
(2) Build each scenario's legislative process from that world.
(3) For each bill, create a deterministic vote context containing party positions, status quo, and seeded randomness.
(4) Pass the bill through the scenario process chain.

(5) Record the bill outcome, including agenda disposition, vote totals, enactment, amendments, challenges, lobbying signals, public-review signals, and policy movement.
(6) Update the scenario's status quo when policy is enacted or rolled back.
(7) Aggregate outcomes into scenario metrics.

Campaigns repeat this over many runs and assumption cases. Scenarios share generated worlds but use independent deterministic random streams, supporting paired comparison among institutions.

5 Design Concepts

5.1 Basic Principles

The model assumes legislative outcomes are shaped by proposal access, agenda scarcity, voting thresholds, committee and information structures, lobbying and capture, amendment opportunities, public review, concentrated harm, reversibility, and strategic actor adaptation.

5.2 Emergence

Productivity, revision moderation, capture, volatility, legitimacy, and gridlock emerge from bill-by-bill interactions. No actor directly optimizes the reported global metrics.

5.3 Adaptation

Proposers adapt over repeated bills. They can reduce proposal volume, delay timing, moderate risky bills, seek outside-bloc cosponsorship, reduce lobby exposure, add compensation amendments, lower risk tolerance after bad outcomes, and withdraw weak risky bills. Proposer state includes trust, proposal pace, risk tolerance, concession rate, lobby exposure, cooldown, and streak counters.

Lobby groups adapt over repeated bills. They track returns by channel, select capture strategies from learned returns and current bill conditions, change issue-specific spending multipliers, adjust overall budget intensity, increase reform-threat sensitivity when anti-lobbying bills become threatening, and maintain defensive pressure against reforms that would reduce lobbying influence.

Institutional state also adapts through proposal credits, proposal bonds, law registries, sunset review, challenge-token exhaustion, attention-token spending, norm erosion, judicial reversals, and active-law correction.

5.4 Objectives

Legislators vote according to weighted incentives. Proposers seek agenda access and enactment while learning to avoid reputational or procedural failure. Lobby groups seek policy movement toward their issue preferences and private gains while resisting anti-lobbying reforms. Institutions impose constraints and incentives whose tradeoffs are measured afterward.

5.5 Learning and Prediction

Learning is deterministic and bounded rather than a full reinforcement-learning procedure. Actors use local outcome signals: enactment, public support, public benefit, capture risk, concentrated harm, salience, private gain, prior failures, and prior channel returns.

Table 2. Detailed generator variables and where they enter the model. This table expands the simplified assumptions table in the main paper.

Component	Variable	Distribution or formula summary	Scenario tuned?	Main use
Legislator ideal point	z_i	faction centers at $\{-s, 0, s\}$, $s = 0.25 + 0.65\rho$, Gaussian spread, clamp $[-1, 1]$	yes	vote kernel, parties
Party and district signals	party, d_i	party bins or party-system profile; $d_i = 0.78z_i + N(0, 0.18 + 0.10\rho)$	yes	party pressure, district diagnostics
Sensitivities	$c_i, \pi_i, \kappa_i, \lambda_i, r_i$	scenario center plus bounded Gaussian noise	yes	revision moderation, party, constituency, lobby, reputation weights
Status quo	q_0	$N(0, 0.18)$, clamp $[-1, 1]$; enacted bills update q_t	no	status-quo-relative voting
Bill position	x_b	proposer-centered $z_p + N(0, 0.22)$, with adversarial shock variants	yes	vote kernel, policy shift
Public support/benefit	p_b, g_b	$g_b = 0.34 + 0.34(1 - \lvert x_b \rvert) - 0.08 private + N(0, 0.18)$; $p_b = g_b + 0.08\ell_b - 0.05 private + N(0, 0.22)$	yes	support, yield, risk
Lobby and private gain	$\ell_b, private_b$	$\ell_b = N(0, L)$, $private_b = 0.72 \max(0, \ell_b) + U(0, 0.20)$, bounded	yes	access, capture, vote
Harm and uncertainty	h_b, σ_b	harm rises with distance, salience, private gain, and noise; uncertainty rises with salience, signal mismatch, and lobbying	yes	harm thresholds, risk metrics

5.6 Stochasticity

World generation, committee composition, citizen panels, bill signals, party-system profiles, and some routing decisions use seeded randomness. Identical seeds reproduce identical outputs.

5.7 Observation

The simulator reports productivity, floor load, enacted support, generated public benefit, cooperation, revision moderation, gridlock, access denial, committee rejection, challenge rate, chamber low-support passage, low-public-support enactment, policy shift, proposer gain, lobby capture, public alignment, anti-lobbying success, private-gain ratio, lobbying spend by channel, public benefit per lobby dollar, amendment rate, amendment movement, amendment overload, poison-pill rate, support gain after amendment, minority harm, concentrated-harm passage, compensation rate, legitimacy, enacted-policy quality, policy yield per submitted proposal, blockage reliance, public support, active-law welfare, implementation delay/capacity/failure risk, nonenforcement and underfunding risk, renewal pressure, reversal rate, correction delay, low-support active-law share, selected-alternative distance, proposer agenda advantage, alternative diversity, status-quo win rate, citizen-review metrics, citizen agenda petition metrics, open-calendar metrics, attention spending, objection-window use, public-will review, district alignment, cosponsorship, bond forfeiture, strategic decoys, proposer-access Gini, vetoes, overrides, chamber conflict, malapportionment, population-seat distortion, committee capture, eligibility exclusion, renewal staggering, recusal, ex ante review, review delay, and quiet-capture risk. The main manuscript uses only the subset of metrics in its metric table; the full observation list is retained here to document implemented outputs for reproduction and future studies.

6 Initialization

`WorldGenerator` creates legislators from party-system and ideology parameters, derives party positions, generates lobby groups by issue domain, chooses a scalar status quo, and generates bills around proposer ideal points. Generated values are bounded by implementation constraints and broad plausibility checks. Party-system profiles include ideological bins, two major parties with minor parties, fragmented multiparty systems, dominant-party systems, and weighted campaign profiles.

7 Input Data and Empirical Boundary Checks

Normal scenario runs use synthetic generated worlds. Empirical flow smoke tests read the tracked benchmark extract at `data/calibration/empirical-benchmarks.csv`. The extract maps empirical sources to simulator metrics and broad benchmark ranges. Named sources include Voteview roll-call data [Lewis et al. 2021], Congress.gov and govinfo bill histories [Library of Congress 2026; United States Government Publishing Office 2020], Comparative Agendas Project topic data [Comparative Agendas Project 2026], U.S. Lobbying Disclosure Act filings [United States Senate Office of Public Records 2026], Center for Effective Lawmaking scores [Center for Effective Lawmaking 2026; Volden and Wiseman 2014], and institutional indicators from V-Dem and QoG-linked sources [Quality of Government Institute 2026; V-Dem Institute 2026].

The three govinfo lifecycle censuses cover every H.R. and S. record in the pinned 116th-, 117th-, and 118th-Congress BILLSTATUS bulk archives: respectively 14,148 bills and 68,345 direct actions; 15,066 and 72,047; then 16,213 and 75,239. Each normalized row stores byte-level source-record and canonical action hashes, archive provenance, stage evidence, and explicitly preserved source-date anomalies (twenty-four, five, and twenty). Operational classifiers distinguish ordered reported from a report action, use an ordered-reported/reported/discharged union for committee advancement, and require substantive action for floor consideration. The GPO guide cautions that action types are processing categories and provides no complete authoritative code list, so these stages are reproducible research classifications rather than official legal-status determinations.

Full 118th processing exposed a context-dependent presidential action code: `E30000` labels signatures in most records but the veto of S.4199 in the Senate-origin stream. Classifier v2 therefore removed `E30000` from code-only enactment and required positive signature/enactment text or an unambiguous law record or code. Classifier v3 adds a successful-override stage only when affirmative House and Senate override evidence is present. Applied symmetrically, the revisions change no established funnel count. The 116th has 334 presentments, 333 enactments, two vetoes, and one successful override (H.R.6395); the 117th has 358 presentments and enactments with no veto; and the 118th has 270 presentments, 269 enactments, and one non-overridden veto. Nine 116th and three 118th bills have separate House and Senate passage evidence but no evidence of agreement on identical text, so the conservative completed-passage flag remains false.

The deterministic split holds out bills whose SHA-256 bill-ID prefix has even parity, leaving 7,564 calibration and 7,502 held-out bills. A fixed 50-seed, 24-runs-per-seed panel evaluates calendar-priority thresholds from 0.60 through 0.74. Selection minimizes a prespecified tolerance-standardized squared error over floor consideration and enactment using only the calibration half. Committee advancement is reported as an upstream workflow check and does not enter calendar-threshold selection. The selected and default threshold, 0.68, produces committee-advance, substantive-floor, and enactment rates of 0.106278, 0.061097, and 0.027417; the corresponding held-out census rates are 0.101440, 0.067715, and 0.024927. All 50 leave-one-seed-out panels reselect 0.68.

The complete 116th and 118th censuses are then used without refitting. The 116th committee-advance, substantive-floor, and enactment rates are 0.105174, 0.074074, and 0.023537; all three pass the frozen absolute-error tolerances of 0.020, 0.015, and 0.010. The corresponding 118th rates are 0.113674, 0.059027, and 0.016592. Committee and floor pass, while enactment's 0.010825 error misses by 0.000825. Thus five of six external cohort-metric cells pass, and all six remain inside broader 117th-derived ranges. The earlier pass means the later enactment miss is not a recurring directional bias across both external cohorts, but two cohorts do not establish general transport.

A separate denominator-aligned executive-action diagnostic prevents aggregate fit from hiding a mechanism error. The compact panel parses 126,760 H.R./S. records from 22 pinned 108th–118th-Congress GovInfo archives and retains 4,021 presidential decisions. Each Congress satisfies presentments equal enactments plus vetoes minus overrides. The panel contains 21 vetoes and six successful overrides, for a pooled conditional veto rate of 0.005223 (95% Wilson interval [0.003419, 0.007971]). The frozen selected simulator panel produces 647 vetoes among 2,621 executive decisions, 0.246852 [0.230724, 0.263722], or an exact-count ratio of 47.266 times the empirical rate; the intervals do not overlap. The exact veto set, dates, kinds, and override outcomes match official Senate histories; president and chamber-control context follows the official House history. Divided-government and opposition-sponsor strata have higher observed rates, but they are selected after congressional passage and do not estimate causal party effects. Joint resolutions are excluded. Because no veto-specific tolerance was prespecified, this remains a descriptive mechanism discrepancy rather than a post-hoc formal pass/fail test. The current veto parameterization is therefore an elevated-propensity stress mechanism, not an empirically calibrated presidential-choice model.

The executable checker runs conventional scenarios and writes CSV and Markdown pass/fail reports under `reports/`. A pass is an empirical screen; only the calendar-capacity threshold described above is selected against the census calibration half. Benchmark ranges remain intentionally broad and several are abstract proxies, so passing should be read as minimum plausibility rather than validation. Other empirical comparison checks include a bounded sponsor aggregate with sponsor-bill metadata matches by Bioguide ID, Voteview roll-call rows carrying bounded member metadata and bill-number linkage, Senate LDA lobbying disclosures with bounded issue-policy context, bounded OpenFEC campaign-finance rows with recipient and limited issue/member/district context, a bounded Cumulative CES district-opinion aggregate [Kuriwaki 2025], a bounded QoG/OWID/V-Dem comparative-institution profile [Bormann and Golder 2022; Henisz 2017; V-Dem and Our World in Data 2026a,b], SCDB merits review with bounded U.S.C.-section authority-overlap metadata, Federal Register final-rule records, and Congress.gov public-law revision text. Fourteen targeted held-out checks across nine source families pass broad ranges. Sponsor-bill metadata, district public-opinion policy-area context, comparative scenario-family anchors, LDA issue-policy context, FEC issue-sector and target-scope context, finance/lobbying source-acquisition targets, Federal Register authority and proposed-history metadata, SCDB section overlaps, topic throughput, and sampled committee summaries remain context or proxy evidence. A separate linkage audit reports all thirteen source families as linked, metadata-linked, or partially linked. Optional API fixtures test adapter shapes but are not validation evidence.

Table 3. Empirical boundary by source family. Held-out benchmark rows have a deterministic held-out empirical check; flow sanity and calibration proxy rows remain limited empirical screens. Full next-source acceptance gates appear in the generated empirical-boundary report.

Source family	Status	Claim boundary
Congress.gov bill histories	calibration proxy (available)	Supports a bounded 118th-Congress Congress.gov to GovInfo source cross-check only; not a bill census model validation public benefit or welfare Linkage: linked.
govinfo bill and action records	held-out benchmark (available)	Supports descriptive census-backed H.R. and S. legislative-flow benchmarks plus deterministic within-Congress checks two no-refit external-Congress tests and an 11-Congress executive-action diagnostic; not causal model validation public opinion public benefit welfare or institutional ranking Linkage: linked.
Voteview roll-call data	held-out benchmark (available)	Supports held-out roll-call coalition benchmark and party-unity plausibility only; not district public opinion representation or public-support validation Linkage: metadata linked.
Comparative Agendas topic throughput	flow sanity check (available)	Supports agenda-flow plausibility; does not validate public value by topic Linkage: linked.
QoG and V-Dem comparative institutions	held-out benchmark (medium)	Supports held-out comparative bicameral-context plausibility and bounded simulator scenario-family metadata anchoring only; does not validate institutional fit bicameral disagreement or law-output productivity Linkage: metadata linked.
Senate LDA lobbying disclosures	calibration proxy (available)	Supports spending observability and concentration only; not bill-level capture validation Linkage: partially linked.
OpenFEC campaign finance	held-out benchmark (high)	Supports held-out campaign-finance concentration and outside-spending plausibility only; not bill-level influence or capture validation Linkage: partially linked.
Center for Effective Lawmaking and sponsor histories	held-out benchmark (available)	Supports held-out sponsor proposal-access concentration benchmarking and bounded sponsor-to-bill metadata linkage only; not full member effectiveness or sponsor-success validation Linkage: metadata linked.
District public opinion and affected groups	held-out benchmark (high)	Supports held-out district survey aggregation and one bounded source-reviewed historical related-issue pilot with two privacy-thresholded annual direct-weighted district estimates; not exact or contemporaneous bill support MRP design-based uncertainty validated boundary equivalence bill-text-specific affected-group mapping causal representation public benefit or model validation Linkage: partially linked.

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Empirical boundary by source family (continued)

Source family	Status	Claim boundary
Committee hearing markup referral and discharge records	flow sanity check (available)	Supports rough committee reporting only; hearings markups and amendments remain thin Linkage: linked.
Court review and invalidation	held-out benchmark (medium)	Supports held-out merits-case court-review plausibility and bounded U.S.C.-section authority-overlap metadata only; emergency-order behavior and direct case-to-public-law review remain unvalidated Linkage: partially linked.
Rulemaking implementation and enforcement	held-out benchmark (high)	Supports held-out final-rule implementation-delay plausibility only; complete comment records, enforcement, nonenforcement, and underfunding remain unvalidated Linkage: metadata linked.
Statutory revision and law lineage	held-out benchmark (high)	Supports held-out statutory revision-activity plausibility bounded official OLRC public-law marker review target review packets and a thirteen-public-law source-reviewed target-section diff pilot only; not full target-statute longitudinal lineage observed expiration outcomes effective statutory text or direct court invalidation Linkage: metadata linked.

Table 4. Vote-score feature functions and weights. The anti-capture kernel changes weights; affected-group variables enter procedural thresholds and metrics rather than the base yes/no vote score.

Feature	Function summary	Standard w_k	Anti-capture w_k
Ideology	$(u_i(x_b) - u_i(q_t))(0.75 + salience_b)$	1.45	1.45
Party	party loyalty times $u_{party}(x_b) - u_{party}(q_t)$	0.85	0.78
Constituency/public	constituency sensitivity times $2(p_b - 0.5)$	0.95	1.12
Lobbying	lobby sensitivity times ℓ_b	0.55	0.14
Revision moderation	revision-moderation preference times 0.45 moderation +0.35 public legitimacy +0.20 incrementalism	0.75	0.82
Reputation	reputation sensitivity times unpopular-bill and lobby-capture penalties	0.55	0.92
Noise	seeded Gaussian perturbation before logistic voting	sd 0.25	sd 0.25

8 Submodels

8.1 Voting

Legislators cast YAY or NAY through weighted influences: ideology, party, constituency/public support, lobbying, revision-moderation preference, reputation, and affected-group sensitivity.

8.2 Proposal Access and Agenda Scarcity

Access rules include open access, open floor calendars with abusive-access screens, citizen agenda petitions, viability screens, scalar proposal costs, public-value waivers, lobby surcharges, member quotas, leadership agenda control, public-interest screens, anti-capture proposal-access screens, cross-bloc cosponsorship gates, affected-group sponsor gates, proposal bonds, earned proposal credits, agenda lotteries, and quadratic attention budgets.

8.3 Committees and Information

Committee gatekeeping can reject proposals before floor consideration. Committee information processes move public-signal estimates toward generated public benefit with composition-dependent accuracy. Configured committee selection can vary issue domain, random seed, party quota, proportional assignment, expertise weighting, independence weighting, leadership control, rotation memory, chair allocation, quorum party balance, citizen advisory weight, and topic-referral authority. Committee-power processes distinguish priority queues, veto players, fast-track certification, scrutiny and audit, public accounts, legal/drafting review, and discharge-petition targets.

8.4 Chamber Structure, Eligibility, and Independent Review

Chamber specifications define equal-population, proportional, mixed-member, territorial, malapportioned, appointed, indirect, functional, or mixed elected/appointed chambers. Bicameral routing variants can change origination, emergency routing, second-chamber preclearance, amendment-only powers, absolute or territorial vetoes, suspensive vetoes, navette, mediation, last-offer bargaining, joint sittings, and lower-house overrides. Eligibility and retention filters model stylized age, residency, language, expertise, conflict, attendance, capture-risk, dual-office, party/executive overlap, recusal, cooling-off, term-limit, recall, public-support retention, and sanctions rules. Ex ante review variants model optional referral, mandatory high-risk review, advisory opinions, suspensive objections, mandatory clearance, qualified override, and deadline-limited review by insulated fiscal, electoral, audit, ethics, cost-estimator, regulator, or boundary institutions.

8.5 Challenge and Objection

Challenge vouchers allocate scarce blocking rights to parties or members. Q-member escalation lets a sufficient number of legislators trigger active review. Public objection windows route contested bills to active voting, higher thresholds, or repeal review depending on scenario.

8.6 Lobbying

Budgeted lobbying modifies bill pressure and public signals through channel-specific spending. Anti-capture variants include transparency, public financing, public advocates, blind review, audits, sanctions, and defensive-spend caps. Adaptive lobbying learns from channel returns and reform threats.

8.7 Amendment and Mediation

Structured mediation moves risky proposals toward weighted institutional targets. Multi-round mediation models repeated amendments, costs, concession limits, poison-pill risk, capacity pressure, public-support improvement against a no-amendment counterfactual, proposer-advantage reduction, and compensation for concentrated harm.

8.8 Distributional Harm

Harm-weighted thresholds, compensation amendments, and affected-group consent mechanisms distinguish broad aggregate benefit from concentrated minority loss.

8.9 Citizen Mini-Publics

Citizen panels simulate sampling noise, information quality, manipulation risk, informed support, benefit estimates, and legitimacy. Panel output can affect burden-shifting eligibility, active-vote routing, final threshold, agenda priority, or affirmative majority burden of proof. Citizen agenda-petition paths separate raw support from costly participation signals, public-support intensity, and cheap signal distortion caused by organized amplification.

8.10 Leadership, Direct Democracy, and Judicial Review

Leadership agenda control schedules bills using observable public support, salience, uncertainty, majority-party fit, capture risk, and concentrated-harm risk rather than generated true welfare. Citizen-initiative processes let sufficiently salient, high-support proposals bypass legislative floor blockage through a public referendum. Conference-committee processes revise near-miss bicameral bills toward a bargaining target before chamber revotes. Judicial-review processes can invalidate enacted laws with high rights-like harm, uncertainty, policy shock, or capture risk.

8.11 Alternatives, Package Bargaining, Norm Erosion, and Law Registry

Competing-alternative processes generate same-domain alternatives and select by public benefit, public support, chamber-median proximity, or pairwise majority before final ratification. Pairwise amendment tournaments use the same selector logic for amended versions, making amendment control an explicit pre-ratification stage rather than only a final yes/no vote. Package-bargaining processes approximate revision through side-payment costs, implementation-delay costs, modest policy movement, and harm-reducing package trades. Norm-erosion processes raise procedural delay and reduce public support signals under high contention. Law-registry processes allow enacted laws to persist,

Table 5. Selected core metrics from the main campaign, grouped by mechanism family. Entries are median [IQR] across broad and adversarial cases; the appendix reports mean/range, yield, seed diagnostics, and separate PAIR/AMT rows.

Label	Representative system	Prod. ↑	Rev. mod. ↑	Public ↑	Low supp. ↓	Risk ctrl. ↑	Admin feas. ↑	Example profile ↑
CUR*	**Stylized U.S.-like benchmark**	**0.05 [0.03,0.06]**	**0.37 [0.36,0.38]**	**0.79 [0.78,0.80]**	**0.01 [0.00,0.01]**	**0.94 [0.92,0.94]**	**0.02 [0.01,0.02]**	**0.57 [0.56,0.58]**
SM	Simple majority	0.27 [0.22,0.32]	0.23 [0.23,0.25]	0.69 [0.67,0.70]	0.19 [0.18,0.23]	0.87 [0.85,0.88]	0.07 [0.07,0.07]	0.54 [0.53,0.55]
COMM	Committee regular order	0.16 [0.13,0.19]	0.29 [0.28,0.29]	0.74 [0.73,0.74]	0.09 [0.07,0.10]	0.91 [0.88,0.92]	0.03 [0.03,0.03]	0.55 [0.54,0.56]
DIS	Discharge petition	0.14 [0.09,0.17]	0.31 [0.30,0.31]	0.76 [0.74,0.76]	0.04 [0.03,0.04]	0.93 [0.92,0.94]	0.08 [0.07,0.08]	0.56 [0.55,0.56]
OPEN	Open floor calendar	0.31 [0.25,0.37]	0.24 [0.23,0.26]	0.67 [0.67,0.68]	0.25 [0.22,0.26]	0.85 [0.84,0.86]	0.10 [0.09,0.10]	0.54 [0.53,0.56]
SEL	Content-selection family	0.54 [0.49,0.56]	0.50 [0.48,0.52]	0.79 [0.77,0.79]	0.00 [0.00,0.00]	0.94 [0.93,0.95]	0.28 [0.28,0.29]	0.67 [0.66,0.69]
ACG	Anti-capture access	0.26 [0.20,0.30]	0.24 [0.23,0.25]	0.71 [0.68,0.71]	0.16 [0.15,0.20]	0.89 [0.88,0.90]	0.06 [0.06,0.07]	0.55 [0.54,0.56]
HARM	Harm-weighted majority	0.25 [0.18,0.27]	0.24 [0.23,0.27]	0.70 [0.68,0.70]	0.19 [0.18,0.23]	0.89 [0.87,0.89]	0.07 [0.07,0.07]	0.54 [0.53,0.55]
DP	Burden-shifting stress case	0.84 [0.84,0.89]	0.11 [0.10,0.13]	0.56 [0.54,0.57]	0.50 [0.47,0.52]	0.62 [0.61,0.63]	0.07 [0.07,0.07]	0.55 [0.54,0.56]
DPM	Burden shift + mediation	0.81 [0.72,0.89]	0.22 [0.21,0.26]	0.61 [0.59,0.64]	0.40 [0.34,0.43]	0.74 [0.70,0.75]	0.21 [0.19,0.22]	0.58 [0.57,0.60]
PORT	Portfolio hybrid	0.51 [0.44,0.58]	0.41 [0.40,0.45]	0.76 [0.76,0.78]	0.03 [0.01,0.04]	0.93 [0.93,0.93]	0.23 [0.23,0.24]	0.64 [0.63,0.66]

Note: Example profile is one revision-moderation-weighted sensitivity check:
0.30 Rev. mod. + 0.20 Prod. + 0.20 Risk ctrl. + 0.15 Admin feas. + 0.15 Public. Admin feas. is the inverse of the administrative-cost index shown in the Admin column. Low supp. is low-public-support enactment. The asterisk marks the stylized U.S.-like conventional benchmark. Cell values carry the comparison; shading follows the column arrow as a secondary cue.

face implementation delay, operate under capacity/nonenforcement/underfunding risk, be renewed under lobbying pressure, and be repealed or partially rolled back after review.

9 Experimental Design

The main submitted evidence comes from the main comparison campaign. That campaign joins broad assumption cases, adversarial proposal-generator cases, party-system sensitivity cases, and a stylized rising-contention timeline in one artifact. The main manuscript focuses on the stylized U.S.-like benchmark, simple majority, committee regular order, discharge petition, open floor calendar, a combined content-selection family, anti-capture access, harm-weighted majority, a small burden-shifting stress-test family, and a portfolio hybrid. The appendix tables report separate pairwise-alternative and amendment-tournament rows plus the broader implemented catalog, including leadership agenda control, chamber and committee variants, citizen petitions and initiatives, mini-publics, agenda scarcity, package bargaining, review and correction systems, adaptive routing, norm erosion, and influence-system safeguards. Adversarial generator cases test high-benefit extreme reform, popular harmful bills, moderate capture, delayed-benefit reform, concentrated rights-like harm, anti-lobbying backlash, revision dilution, useful lobbying information, and public-opinion error. Follow-up diagnostics include mechanism ablations, manipulation-stress campaigns, chamber-structure family/stress screens, and empirical boundary checks that map optional raw empirical summaries to simulator proxy metrics.

The dense full scenario-average table is not repeated in the PDF appendix. Full numeric outputs, including separate pairwise-alternative and amendment-tournament rows, are provided in `reports/simulation-campaign-v21-paper.csv`; the generated LaTeX full table remains in `paper/figures/scenario_averages_full_table.tex` for local inspection.

The following sensitivity and fairness probes are not validation evidence. They test whether the framework exposes dependence on generator assumptions and whether major qualitative patterns survive limited alternative worlds.

10 Chamber-Structure Supplement

The chamber-structure campaign tests representation architecture, committee power, eligibility filters, and ex ante review separately from the main mechanism comparison.

Table 6. Appendix generator-sensitivity and fairness-control details. These rows summarize robustness probes; they are not validation evidence.

Type	Check	Result	Interpretation
Generator	Extreme-beneficial reform	top PORT 0.72; SEL 0.69; DPM 0.58; DP low supp. 0.57	Tests whether high-distance proposals can carry high generated public benefit.
Generator	Revision dilution	top PORT 0.71; SEL 0.67; DPM 0.57; DP low supp. 0.67	Tests whether moderation can reduce generated public benefit.
Generator	Lobby information	top SEL 0.74; PORT 0.71; DPM 0.65; DP low supp. 0.28	Tests useful organized information rather than only capture pressure.
Generator	Public-opinion error	top PORT 0.68; SEL 0.68; DPM 0.62; DP low supp. 0.41	Tests noisy or systematically biased public-support signals.
Generator	Minority-rights harm	top SEL 0.66; PORT 0.64; DPM 0.63; DP low supp. 0.19	Tests majority support paired with concentrated rights-like harm.
Generator	Technocratic delay	top PORT 0.69; SEL 0.66; DPM 0.61; DP low supp. 0.75	Tests low initial support with delayed generated public benefit.
Fairness	Conventional revision	CUR P 0.05, M 0.36, R 0.93, A 0.02; PROC P 0.05, M 0.32, R 0.93, A 0.03	Gives the conventional family conference/review content improvement.
Fairness	Majority mediation	SM P 0.33, M 0.24, R 0.85, A 0.07; SMED P 0.40, M 0.22, R 0.80, A 0.19	Compares simple majority with a mediation content-improvement step.
Fairness	Committee amendment	COMM P 0.21, M 0.28, R 0.89, A 0.03; CAMD P 0.12, M 0.32, R 0.93, A 0.07	Compares committee gatekeeping with committee information/amendment power.

Note: P is productivity, M is revision moderation, R is risk control, and A is administrative cost. Fairness rows compare content-improvement variants; they do not impose a hard cost-constrained reoptimization.

Table 7. Objective-set frontier sensitivity.

Objective set	Frontier members	Interpretation
Productivity + revision moderation + risk	DP, DPM, SEL	Original narrow frontier.
Productivity + revision moderation + risk + admin	DP, DPM, SEL, PORT, OPEN, SM, HARM, ACG, COMM, DIS, CUR	Adding administrative feasibility makes most selected systems non-dominated.
Yield + risk + admin	DP, DPM, SEL, PORT, OPEN, SM, ACG, COMM, DIS, CUR	Prioritizes generated public benefit per submitted proposal.
Public support + harm + admin	SEL, PORT, DIS, CUR	Prioritizes support, affected-group protection, and feasibility.
Low-support avoidance + productivity	DP, DPM, SEL	Makes low-public-support avoidance a hard objective.

Note: This table is a sensitivity diagnostic. Frontier membership depends strongly on the objective set.

Table 8. Ablation diagnostics for selected mechanisms.

Mechanism	Removed or replaced component	Directional	Rev. mod.	Low supp. reduction	Admin added
Policy tournament	No alternatives	+0.056	+0.266	+0.217	+0.219
Risk routing	No jury lane	-0.002	+0.000	+0.001	+0.008
Anti-capture	Screen only	+0.024	+0.000	+0.043	+0.005
Package bargaining	Scalar package	-0.015	-0.002	+0.004	+0.065
Reversibility	No registry	-0.009	-0.009	-0.034	+0.044

Note: Positive values indicate how much the original mechanism improved relative to the ablated variant, except Admin added, where positive means additional procedural cost in the original mechanism.

Table 9. Manipulation-stress losses for selected mechanisms.

Stress case	Attack or perturbation	Directional loss	Rev. mod. loss	Low supp. added	Admin added
Tournament attack	Clones/decoys	+0.087	+0.046	+0.010	-0.077
Citizen-panel manipulation	Noisy panel	-0.002	+0.018	-0.005	+0.000
Bad-faith harm claims	Loose claims	-0.002	-0.018	-0.018	+0.000
Objection astroturf	Astroturf/noise	+0.012	-0.006	+0.003	+0.010
Agenda flooding	Proposal flood	+0.008	-0.011	+0.007	-0.001

Note: Positive values are losses or added burden under bounded manipulation stress. Low supp. is low-public-support enactment.

Table 10. Representative-versus-family-champion audit. Scores are from the fixed supplemental family-screen baseline, not the main comparison campaign. Negative gaps show cases where the paper uses a readable representative rather than the within-family top scorer.

Label	Family	Family-screen champion	Rep. dir.	Champ. dir.	Gap
CUR	Conventional benchmark	Bicameral upper revision council	0.623	0.633	-0.011
DIS	Agenda gate	Majority + public-interest screen	0.627	0.638	-0.011
OPEN	Other	Long-horizon strategic learning	0.634	0.660	-0.026
DPM	Default-pass side family	Default pass unless 2/3 block	0.688	0.724	-0.036

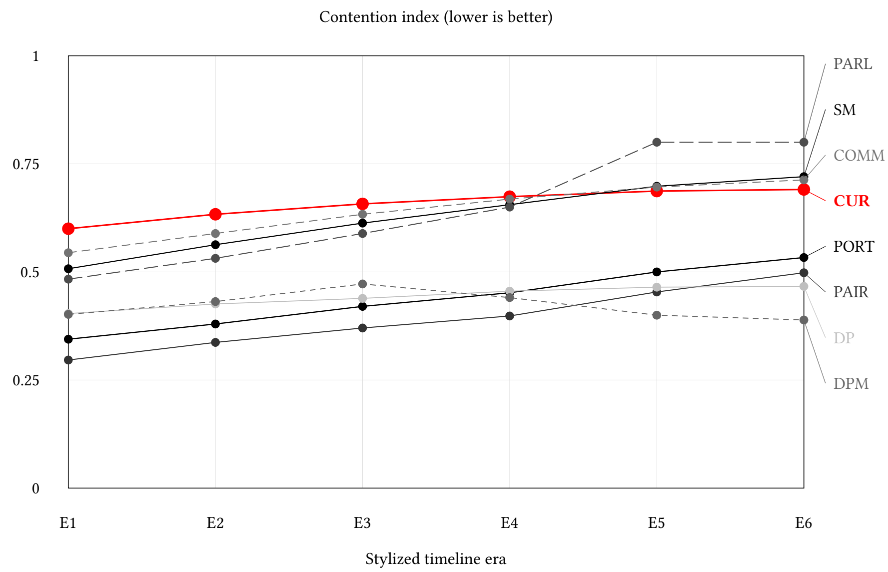

Fig. 1. Appendix rising-contention stress paths from the main comparison campaign. The index is 0.50·gridlock +0.30 · (1−revision moderation) + 0.20·low-public-support enactment. Lower is better.

Table 11. Chamber, apportionment, committee, and review family champions from the chamber-structure supplement. Values are averaged across the chamber campaign cases.

Label	Family champion	Prod.	Rev. mod.	Risk	Malapp.
BICAM	Bicameral conflict	0.32	0.26	0.86	0.00
ASSIGN	Committee assignment	0.23	0.28	0.89	0.00
COMMP	Committee power	0.26	0.27	0.87	0.00
CONV	Conventional benchmark	0.34	0.24	0.84	0.00
REVIEW	Independent review	0.34	0.24	0.84	0.00
ROUTE	Origination and routing	0.32	0.26	0.85	0.00
OTHER	Other chamber design	0.31	0.26	0.85	0.00
SEAT	Seat allocation	0.34	0.24	0.84	0.00
SELECT	Selection and retention	0.36	0.24	0.83	0.00
UPPER	Upper-chamber composition	0.36	0.26	0.84	0.12

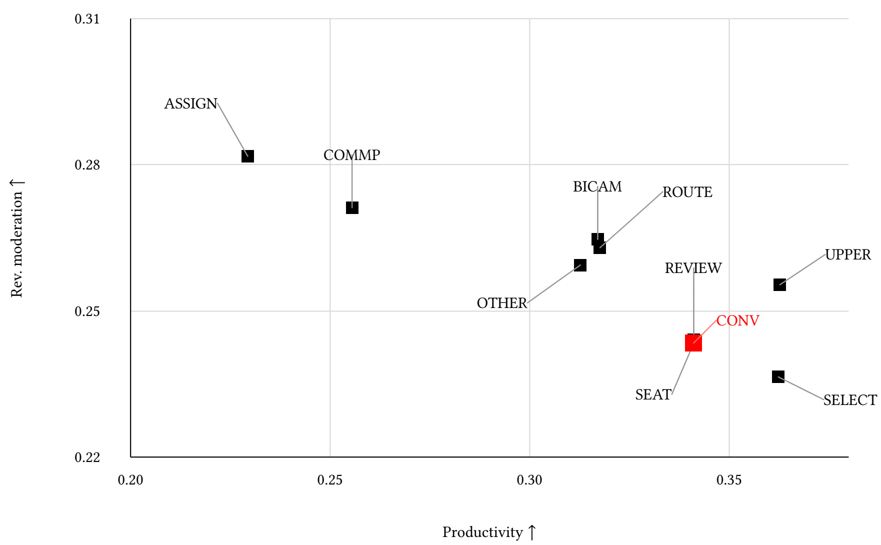

Fig. 2. Chamber-structure family champions from the supplemental chamber campaign, zoomed to the observed champion range so clustered family labels remain readable. The axes keep the paper's central productivity/revision-moderation comparison visible; malapportionment remains in Table 11 because the champion set has little variation on that metric.

11 Assumptions and Limitations

The policy space is one-dimensional. Public benefit is generated rather than empirically estimated. Legislators are synthetic and do not represent named real officials. Lobby groups are abstract collective actors. The empirical boundary layer screens plausible ranges and narrow held-out slices but does not fit raw datasets automatically. Leadership, initiative, conference, judicial, court-architecture, district-public, influence-system, long-horizon learning, and norm-erosion processes are stylized mechanism prototypes rather than detailed institutional models. Institutional mechanisms are simplified so they can be compared in bundles without modeling the full legal, administrative, judicial, media, and electoral environment. Omnibus and package bargaining are proxies for side payments, delays, exemptions, jurisdictional trades, distributive load, and harm reduction, not full models of legal text, appropriations, or multidimensional coalition formation.

12 Reproducibility

The main reproducibility entrypoints are:

- `make test`: compile and run the simulator test suite.
- `make campaign` or `make paper-campaign`: regenerate the comparison campaign.
- `make calibrate`: run empirical flow benchmark screening.
- `make seed-robustness-check`: run multi-seed checks for paper scenarios.
- `make validation-readiness`: check optional raw-input shape.
- `make empirical-validation`: summarize any present raw inputs.

- `make empirical-bridge`: map raw summaries to simulator proxies for empirical flow smoke tests.
- `make empirical-flow-heldout`: run narrow held-out source-family benchmark checks.
- `make govinfo-bill-census-116`: regenerate the complete 116th-Congress H.R./S. external-cohort summary.
- `make govinfo-bill-census`: regenerate the full 117th-Congress H.R./S. lifecycle summary from the committed census.
- `make govinfo-bill-census-118`: regenerate the complete 118th-Congress H.R./S. temporal-cohort summary.
- `make govinfo-executive-action-panel`: verify the compact 11-Congress decision panel offline.
- `make legislative-lifecycle-calibration`: rerun the fixed current-Congress workflow threshold sweep.
- `make legislative-executive-action-diagnostic`: compare empirical presentment, veto, override, and enactment pathways with the frozen simulator panel.
- `make legislative-lifecycle-temporal-replication`: apply the frozen selected threshold to the 116th- and 118th-Congress rates without refitting.
- `make govinfo-bill-census-check`: verify all three censuses, 22 executive-panel archive pins, classifier edge cases, official veto references, cross-source agreement, calibration linkage, the 5 / 6 transport outcome, and the executive-action discrepancy.
- `make empirical-linkage-report`: audit explicit raw-data joins across source families.
- `make empirical-linkage-roadmap`: write join-key requirements and acceptance gates for non-fully-linked source families.
- `make govinfo-billstatus-linkage`: write the bounded govinfo BILLSTATUS cross-check for the cached bill-progression sample.
- `make court-law-linkage`: write the bounded SCDB/Federal Register U.S.C.-section authority-overlap report.
- `make rulemaking-authority-linkage`: write the bounded Federal Register public-law authority-search report.
- `make rulemaking-history-linkage`: write the bounded Federal Register proposed-rule history report for authority-matched final rules.
- `make bill-law-evidence-spine`: write the bounded public-law bill/action join inventory for revision-text proxies, sponsor-district public-opinion context, authority-search matches, proposed-history/comment-portal metadata, final-rule/proposed-to-final timing metadata, and court U.S.C.-section overlaps.
- `make district-public-opinion-policy-context`: write the bounded sponsor-district public-law bill policy-area context inventory for CES district proxy rows.
- `make campaign-finance-district-context`: write the bounded FEC House-candidate district-context inventory.
- `make campaign-finance-member-context`: write the bounded FEC candidate-to-Voteview member-context inventory.
- `make campaign-finance-issue-context`: write the bounded FEC transaction-label issue-context inventory.
- `make validation-gap-report`: generate the paper-facing empirical-boundary report. Adapter fixtures live separately and do not count as raw empirical data.
- `make build-bill-progression-raw`: rebuild the documented Congress.gov bill-progression raw sample when a Congress.gov API key is available.
- `make build-govinfo-bill-census-116-raw`: rebuild the pinned 116th-Congress H.R./S. external cohort from the official bulk archives.

- `make build-govinfo-bill-census-raw`: rebuild the pinned 117th-Congress H.R./S. lifecycle census from the official bulk archives.
- `make build-govinfo-bill-census-118-raw`: rebuild the pinned 118th-Congress H.R./S. external cohort from the official bulk archives.
- `make build-govinfo-billstatus-linkage-raw`: rebuild the bounded govinfo BILLSTATUS cross-check when network access is available.
- `make build-core-raw-validation`: rebuild the broader Voteview, Congress.gov-derived, and Senate LDA empirical comparison samples when network access and any required API key are available.
- `make build-district-public-opinion-raw`: rebuild the bounded Cumulative CES district-opinion extract when network access and the optional Feather parser dependency are available.
- `make ablation-analysis`, `make manipulation-stress`, and `make mechanism-diagnostics`: run mechanism ablations, manipulation stress tests, and the generated diagnostics table workflow.
- `make family-screen`, `make catalog-breadth`, and `make chamber-structure`: run breadth-catalog and chamber-structure diagnostics.
- `make paper`: build the paper and appendix PDFs.
- `make reproduce-paper-offline`: run the no-network paper reproduction path from fixed seeds and local summaries.
- `make supplement-anonymous`: build the double-blind supplement package.
- `make public-provenance`: after review, generate a non-anonymous provenance manifest.

The review package includes generated campaign reports and PDFs. Public repository identifying details are withheld during double-blind review and can be restored after review if permitted.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. OpenAI ChatGPT/Codex tools available in the authoring environment during the April–May 2026 revision period were used for grammar and style editing, restructuring suggestions, code debugging assistance, and drafting assistance for portions of the simulator implementation, generated-report scripts, and manuscript prose. LLM tools were not used as authors. Tool outputs were edited and checked by the author; references, code outputs, experiments, tables, figures, and claims were manually reviewed against the repository artifacts before inclusion. No large language model is listed as an author.

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